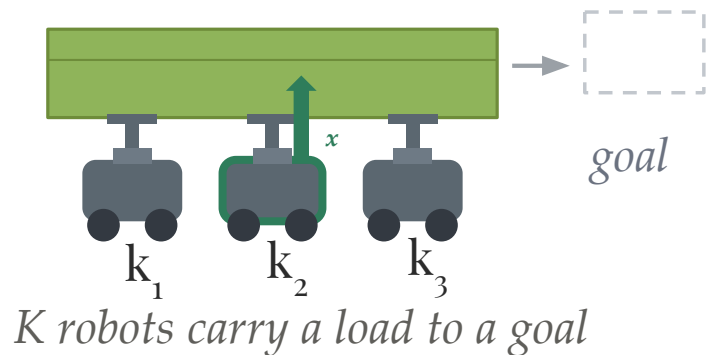


(a) Example Hybrid Action Tasks

Cooperative Transport



Articulated Object Manipulation



Discrete mode + continuous parameters

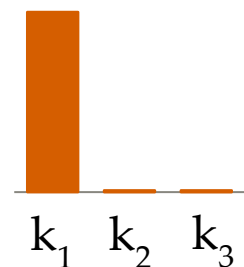
■ *fixture / wall* ■ *movable part* ● *joint axis*

action $a = (m, x)$

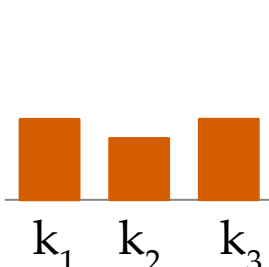
(b) Diffusion vs BFN on Discrete Actions

Diffusion: Relaxes Discrete to Continuous

one-hot target

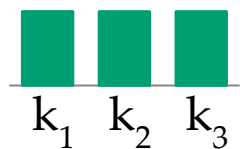
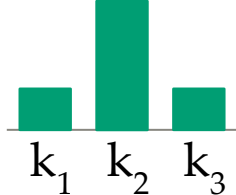
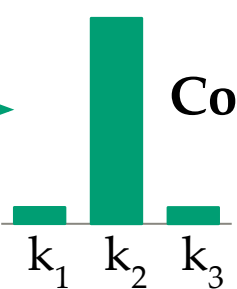
+ Gaussian
noise

noisy output



Error!

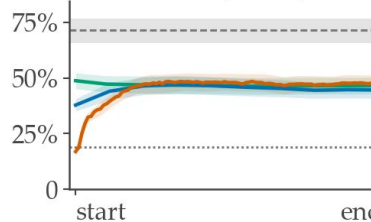
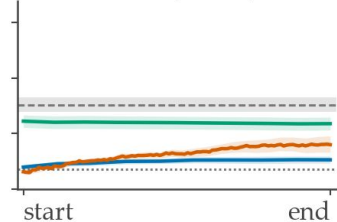
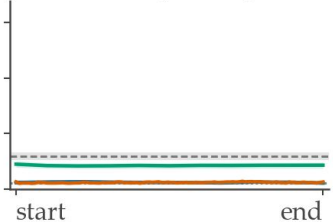
BFN: Refines a Categorical Belief on the Simplex

 $t = 0$ *uniform* $t = 1/2$  $t = 1$ *peaked*

Correct!

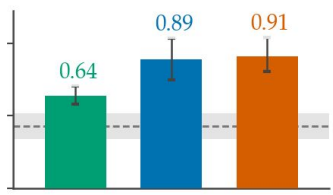
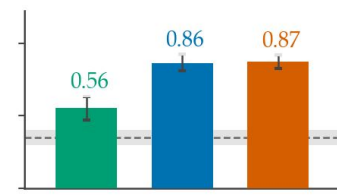
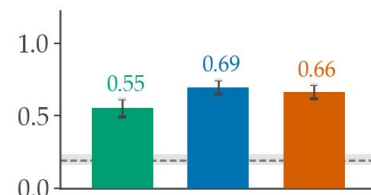
(c) Direct Diagnostics of Discrete-continuous Coupling

Discrete Action Right

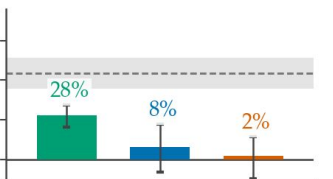
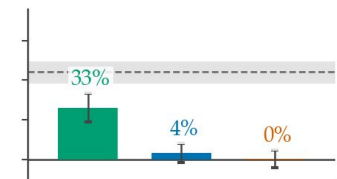
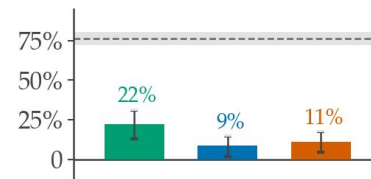
 $n = 4$ ($K = 16$) $n = 6$ ($K = 64$) $n = 8$ ($K = 256$)

Sampling Process

Magnitude Error, Discrete Action Right



Error Lower When Discrete Action Right



BFN-10

DDIM-10

DDPM-100

--- expert (DLPA)

..... random

— 0 = independent